

Polynomial Functions

Definition:

A polynomial function is a function of the form

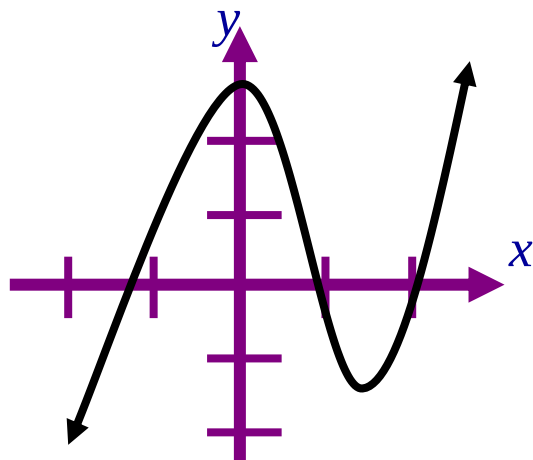
$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where n is a nonnegative integer and each a_i is a real number.
The polynomial function has a leading coefficient a_n and degree n .

Polynomial Function	Leading Coefficient	Degree
$f(x) = -2x^5 + 3x^3 - 5x + 1$	-2	5
$f(x) = x^3 + 6x^2 - x + 7$	1	3
$f(x) = 14$	14	0

Graphs of polynomial functions are continuous.
That is, they have no breaks, holes, or gaps.

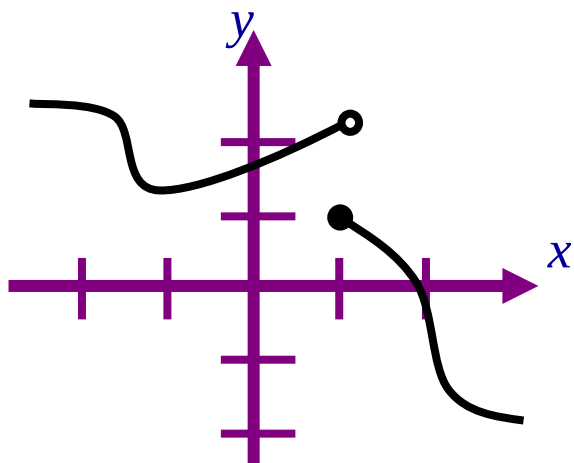
$$f(x) = x^3 - 5x^2 + 4x + 4$$



continuous

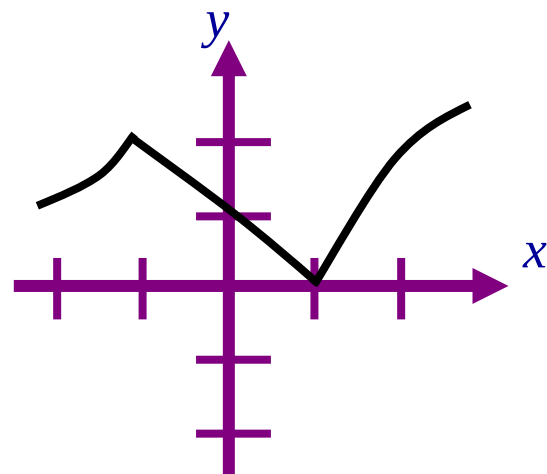
smooth

polynomial



not continuous

not polynomial



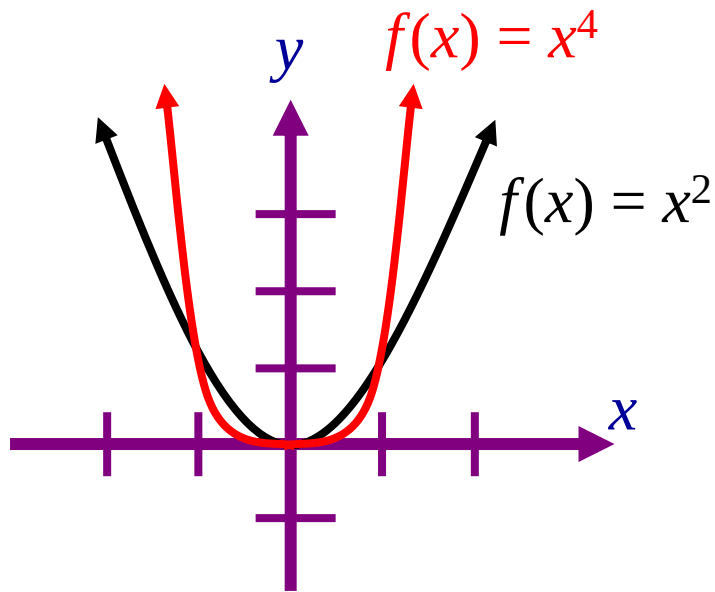
continuous

not smooth

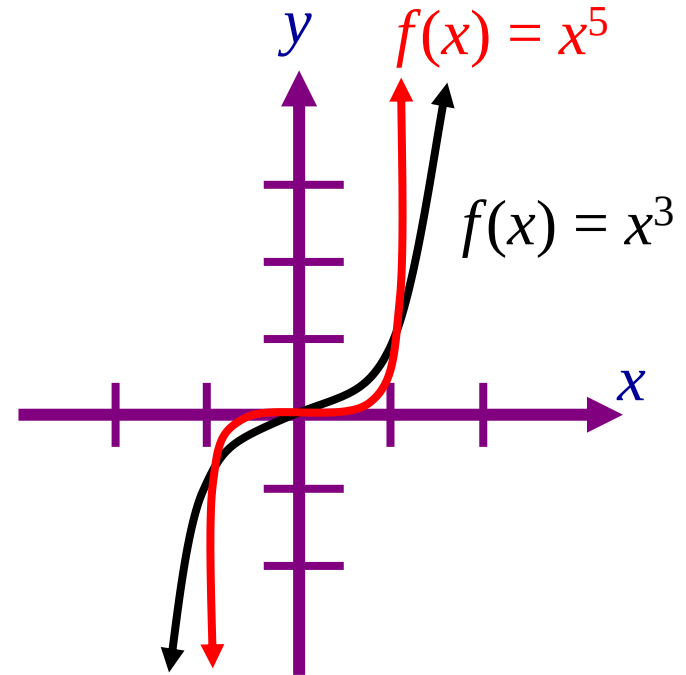
not polynomial

Polynomial functions are also smooth with rounded turns.
Graphs with points or cusps are not graphs of polynomial functions.

Polynomial functions of the form $f(x) = x^n$, $n \geq 1$ are called power functions.



If n is *even*, their graphs resemble the graph of $f(x) = x^2$.

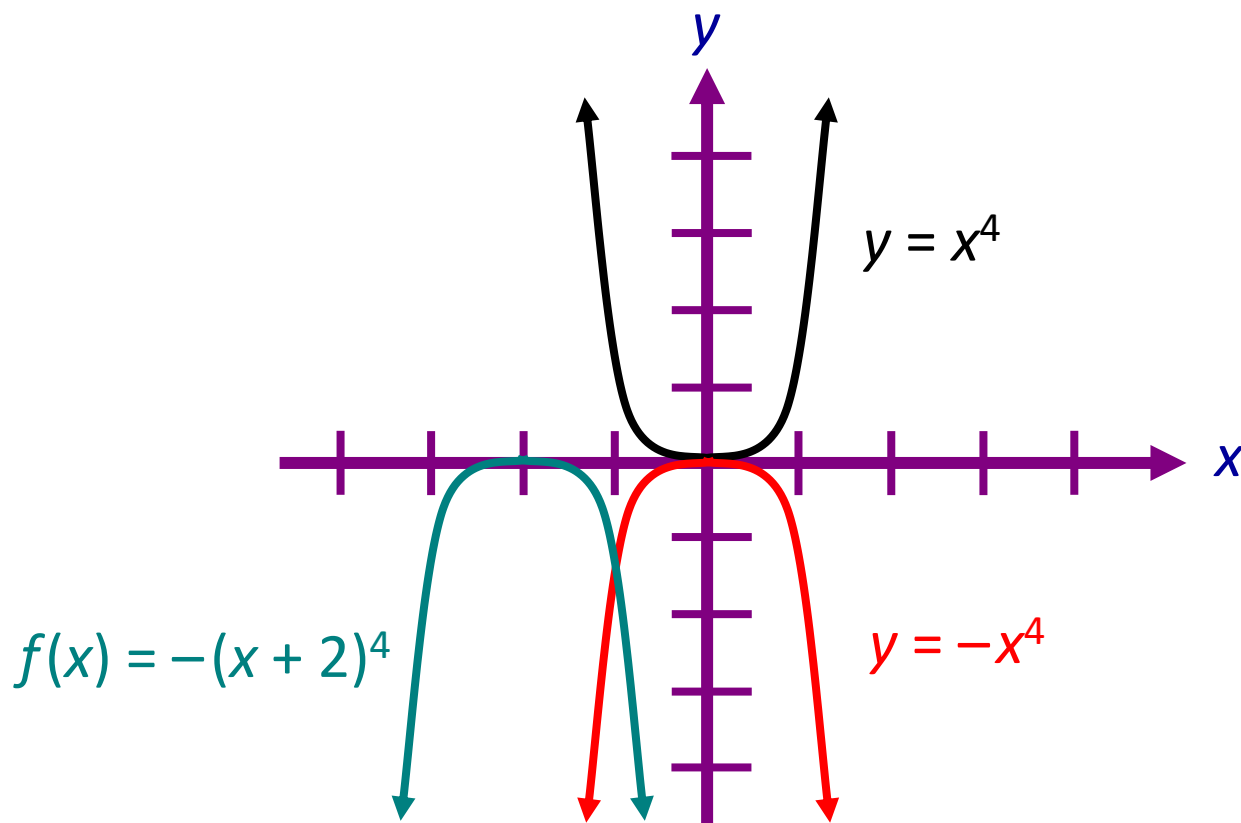


If n is *odd*, their graphs resemble the graph of $f(x) = x^3$.

Example: Sketch the graph of $f(x) = -(x + 2)^4$

This is a shift of the graph of $y = -x^4$ two units to the left.

This, in turn, is the reflection of the graph of $y = x^4$ in the x -axis.

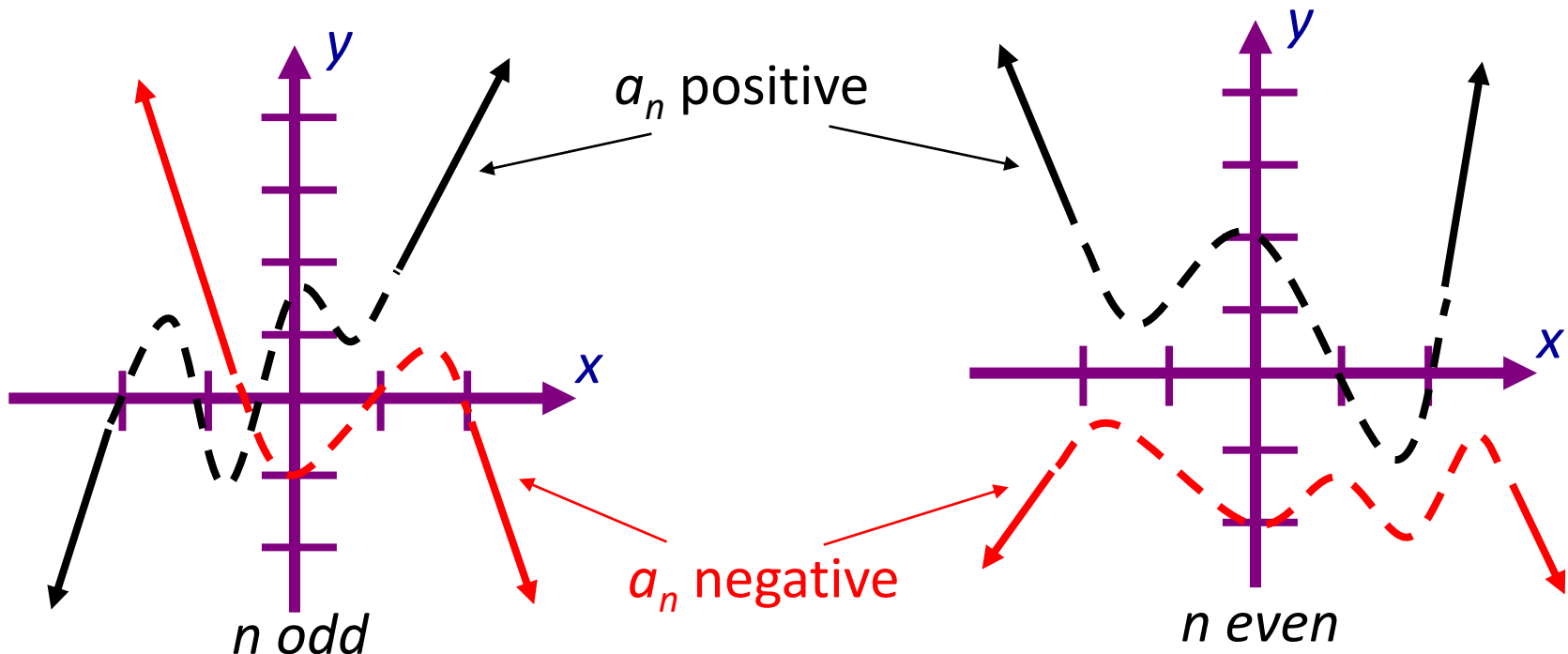


Leading Coefficient Test

As x grows positively or negatively without bound, the value $f(x)$ of the polynomial function

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \quad (a_n \neq 0)$$

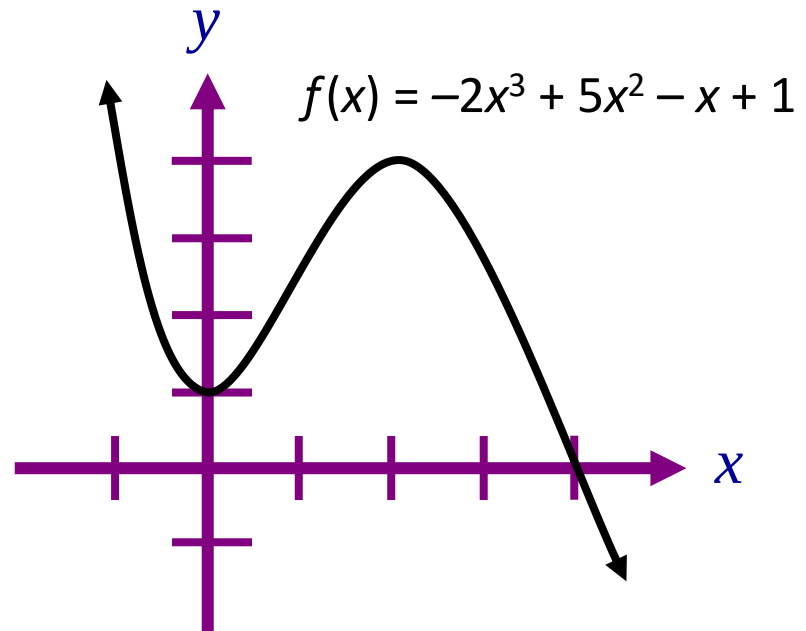
grows positively or negatively without bound depending upon the *sign* of the leading coefficient a_n and whether the degree n is *odd* or *even*.



Example: Describe the right-hand and left-hand behaviour for the graph of $f(x) = -2x^3 + 5x^2 - x + 1$.

Degree	3	Odd
Leading Coefficient	-2	Negative

$$\begin{aligned}x \rightarrow +\infty & \quad f(x) \rightarrow -\infty \\x \rightarrow -\infty & \quad f(x) \rightarrow +\infty\end{aligned}$$



A real number a is a zero of a function $y = f(x)$ if and only if $f(a) = 0$.

Real Zeros of Polynomial Functions

If $y = f(x)$ is a polynomial function and a is a real number then the following statements are equivalent.

1. a is a zero of f .
2. a is a solution of the polynomial equation $f(x) = 0$.
3. $x - a$ is a factor of the polynomial $f(x)$.
4. $(a, 0)$ is an x -intercept of the graph of $y = f(x)$.

A turning point of a graph of a function is a point at which the graph changes from increasing to decreasing or *vice versa*.

A polynomial function of degree n has at most $n - 1$ *turning points* and at most n *zeros*.

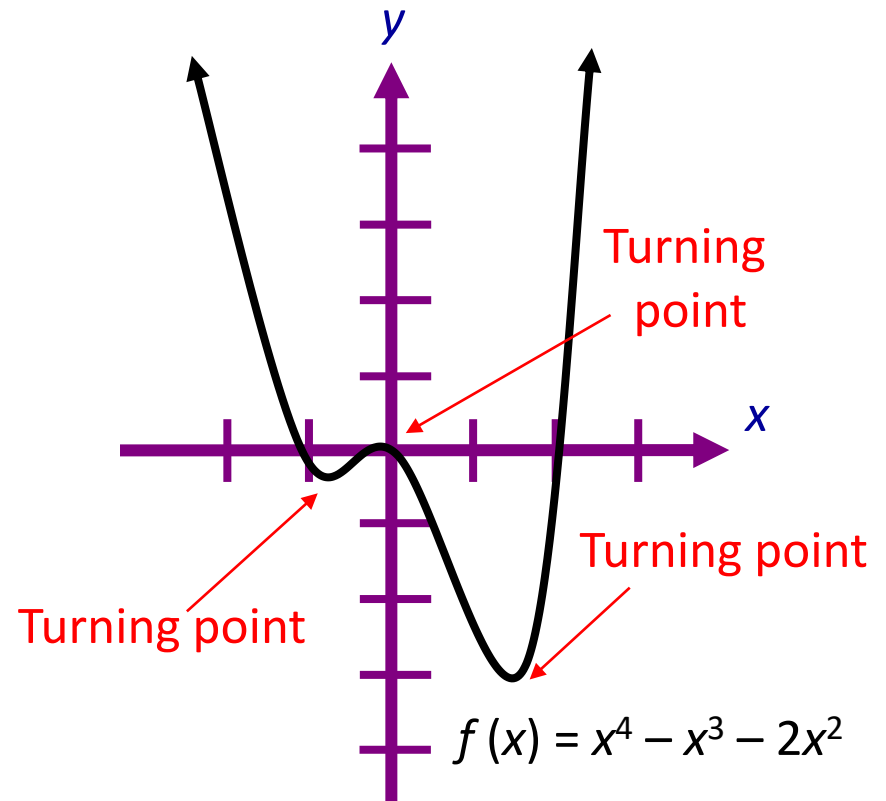
Example: Find all the real zeros and turning points of the graph of $f(x) = x^4 - x^3 - 2x^2$.

Factor completely: $f(x) = x^4 - x^3 - 2x^2 = x^2(x + 1)(x - 2)$

The real zeros are $x = -1$, $x = 0$, and $x = 2$

These correspond to the x-intercepts $(-1, 0)$, $(0, 0)$ and $(2, 0)$.

The graph shows that there are three turning points. Since the degree is four, this is the maximum number possible.



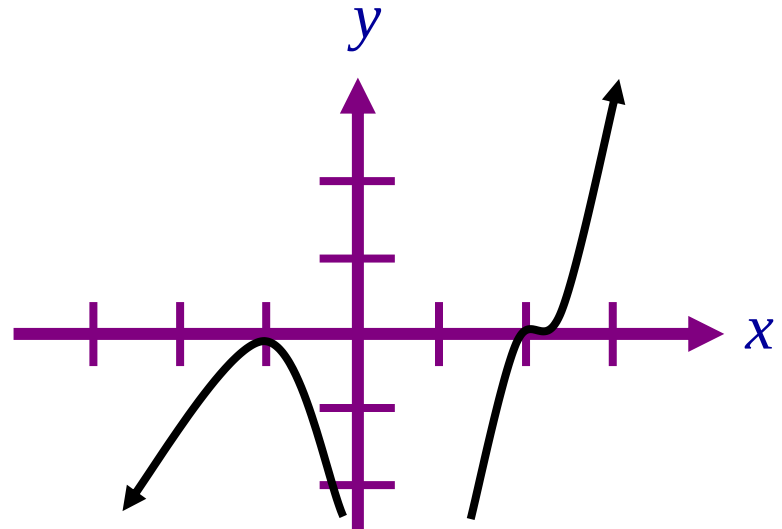
Repeated Zeros of Polynomial Functions

If k is the largest integer for which $(x - a)^k$ is a factor of $f(x)$ and $k > 1$, then a is a repeated zero of multiplicity k .

1. If k is *odd* the graph of $f(x)$ crosses the x -axis at $(a, 0)$.
2. If k is *even* the graph of $f(x)$ touches, but does not cross through, the x -axis at $(a, 0)$.

Example: Determine the multiplicity of the zeros of $f(x) = (x - 2)^3(x + 1)^4$.

Zero	Multiplicity	Behavior
2	3 odd	crosses x -axis at $(2, 0)$
-1	4 even	touches x -axis at $(-1, 0)$



Example: Sketch the graph of $f(x) = 4x^2 - x^4$.

1. Write the polynomial function in standard form: $f(x) = -x^4 + 4x^2$
The leading coefficient is *negative* and the degree is *even*.

as $x \rightarrow \pm\infty, f(x) \rightarrow -\infty$

2. Find the zeros of the polynomial by factoring.

$$f(x) = -x^4 + 4x^2 = -x^2(x^2 - 4) = -x^2(x + 2)(x - 2)$$

zeros:

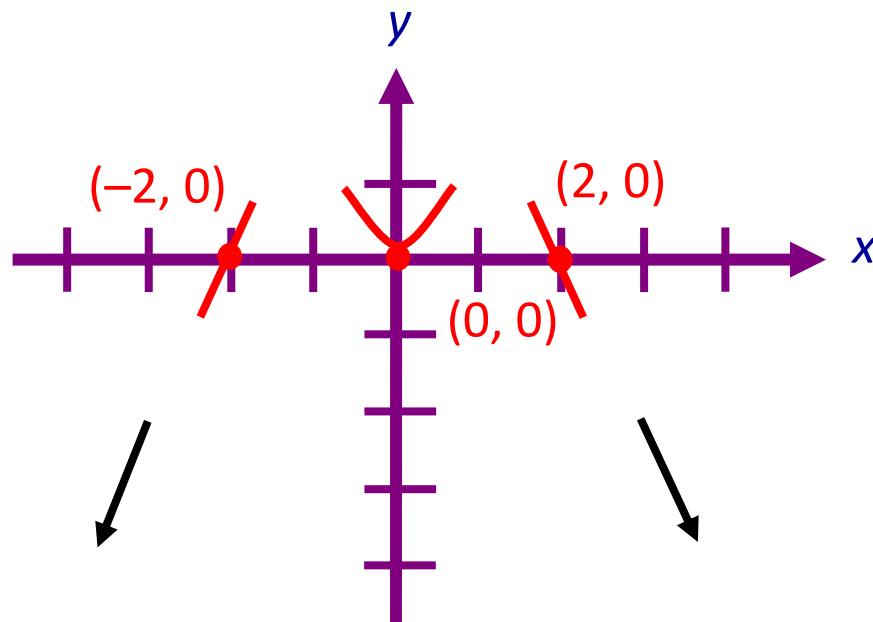
$x = -2, 2$ multiplicity 1

$x = 0$ multiplicity 2

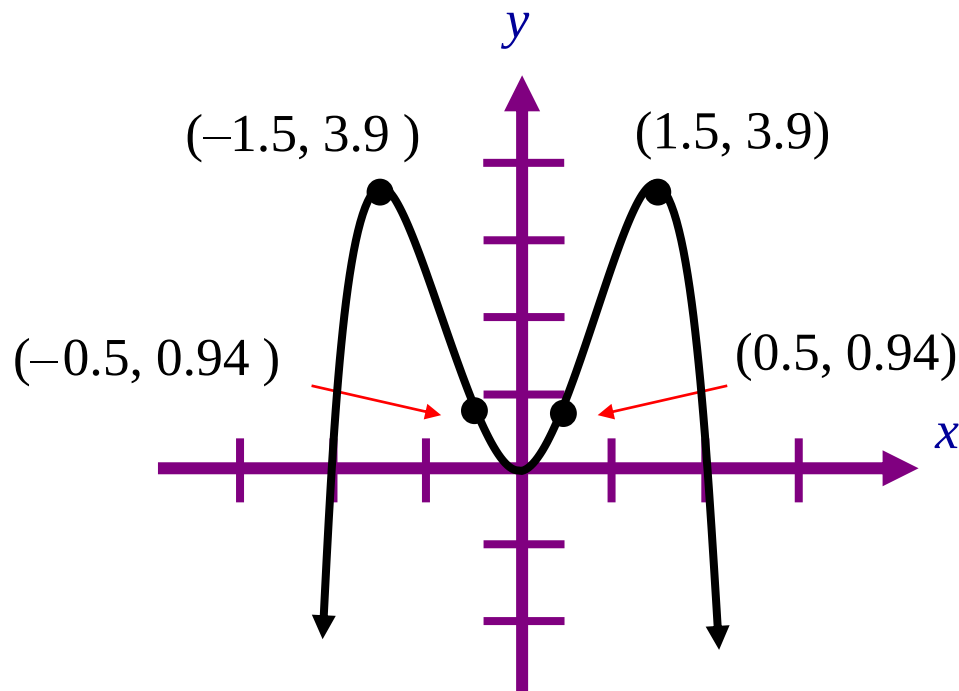
x-intercepts:

$(-2, 0), (2, 0)$ crosses trough

$(0, 0)$ touches only



3. Since $f(-x) = 4(-x)^2 - (-x)^4 = 4x^2 - x^4 = f(x)$, the graph is symmetrical about the y -axis.
4. Plot additional points and their reflections in the y -axis:
 $(1.5, 3.9)$ and $(-1.5, 3.9)$, $(0.5, 0.94)$ and $(-0.5, 0.94)$
5. Draw the graph.



Sketching Curves

First derivative:

y' is positive $\longrightarrow$ Curve is rising.

y' is negative $\longrightarrow$ Curve is falling.

y' is zero $\longrightarrow$ Possible local maximum or minimum.

Second derivative:

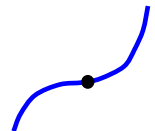
y'' is positive $\longrightarrow$ Curve is concave up.



y'' is negative $\longrightarrow$ Curve is concave down.



y'' is zero $\longrightarrow$ Possible inflection point (where concavity changes).



Example:

$$\text{Graph } y = x^3 - 3x^2 + 4 = (x+1)(x-2)^2$$

There are roots at $x = -1$ and $x = 2$

$$y' = 3x^2 - 6x$$

Possible extreme at $x = 0, 2$

We can use a chart to organize our thoughts

Set $y' = 0$

First derivative test:

$$0 = 3x^2 - 6x$$

$$\begin{array}{ccccccc} y' & \begin{array}{c} + \\ \star \end{array} & 0 & \begin{array}{c} - \\ \star \end{array} & 0 & + & \\ & \hline x & & 0 & & 2 & & \end{array}$$

$$0 = x^2 - 2x$$

$$0 = x(x-2)$$

$$y'(1) = 3 \cdot 1^2 - 6 \cdot 1 = -3 \quad \longrightarrow \text{negative}$$

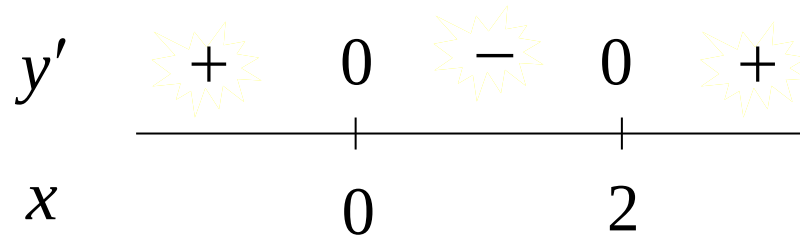
$$x = 0, 2$$

$$y'(-1) = 3(-1)^2 - 6(-1) = 9 \quad \longrightarrow \text{positive}$$

$$y'(3) = 3 \cdot 3^2 - 6 \cdot 3 = 9 \quad \longrightarrow \text{positive}$$

First derivative test:

$$y' = 3x^2 - 6x$$



There is a local maximum at $(0,4)$ because $y' > 0$ for all x in $(-\infty, 0)$ and $y' < 0$ for all x in $(0, 2)$.

There is a local minimum at $(2,0)$ because $y' > 0$ for all x in $(2, \infty)$ and $y' < 0$ for all x in $(0, 2)$.

There are roots at $x = -1$ and $x = 2$

$$y' = 3x^2 - 6x$$

Possible extreme at $x = 0, 2$

Or you could use the second derivative test:

$$y'' = 6x - 6$$

$$y''(0) = 6 \cdot 0 - 6 = -6$$

Because the second derivative at $x = 0$ is negative, the graph is concave down and therefore $(0,4)$ is a local maximum.

$$y''(2) = 6 \cdot 2 - 6 = 6$$

Because the second derivative at $x = 2$ is positive, the graph is concave up and therefore $(2,0)$ is a local minimum.

We then look for inflection points by setting the second derivative equal to zero.

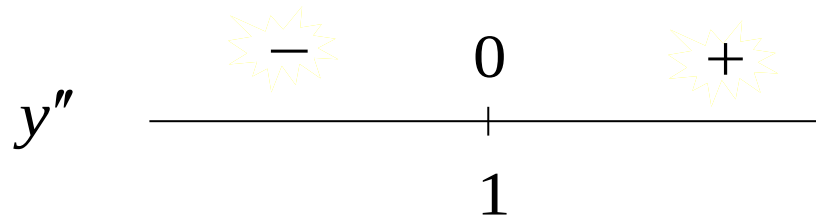
Possible inflection point at $x = 1$

$$y'' = 6x - 6$$

$$0 = 6x - 6$$

$$6 = 6x$$

$$1 = x$$



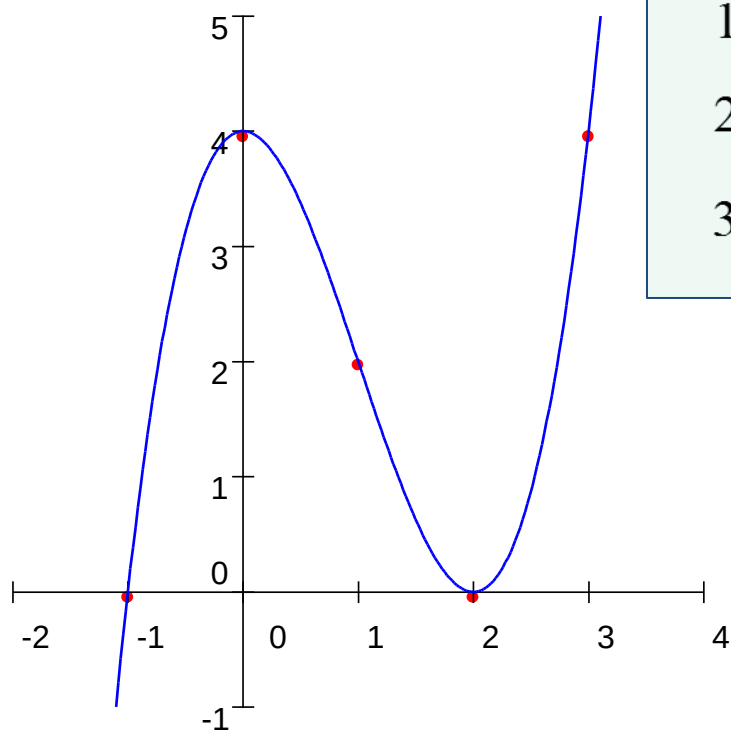
$$y''(0) = 6 \cdot 0 - 6 = -6 \quad \longrightarrow \text{negative}$$

$$y''(2) = 6 \cdot 2 - 6 = 6 \quad \longrightarrow \text{positive}$$

There is an inflection point at $x = 1$ because the second derivative changes from negative to positive.

$\therefore$ inflection point at $x = 1$

Make a summary table:



x	y	y'	y''	
-1	0	9	-12	rising, concave down
0	4	0	-6	local max
1	2	-3	0	falling, inflection point
2	0	0	6	local min
3	4	9	12	rising, concave up

Rational Functions

Definition:

A rational function is a function of the form $f(x) = \frac{P(x)}{Q(x)}$

where $P(x)$ and $Q(x)$ are polynomials and $f(x) \neq 0$.

Example: $f(x) = \frac{1}{x}$ is defined for all real numbers except $x = 0$

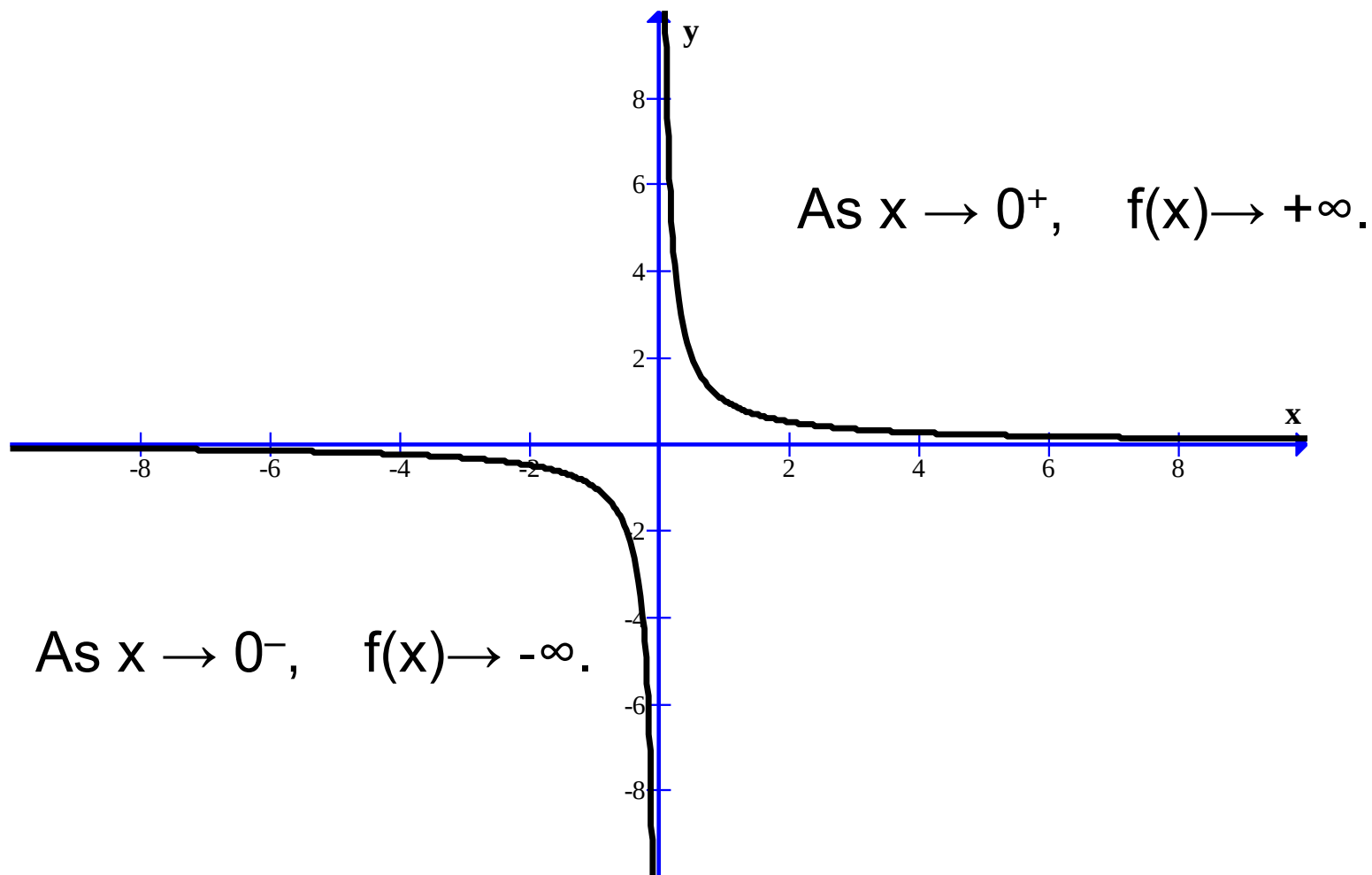
$$f(x) = \frac{1}{x}$$

x	$f(x)$
2	0.5
1	1
0.5	2
0.1	10
0.01	100
0.001	1000

as $x \rightarrow 0^+$, $f(x) \rightarrow +\infty$.

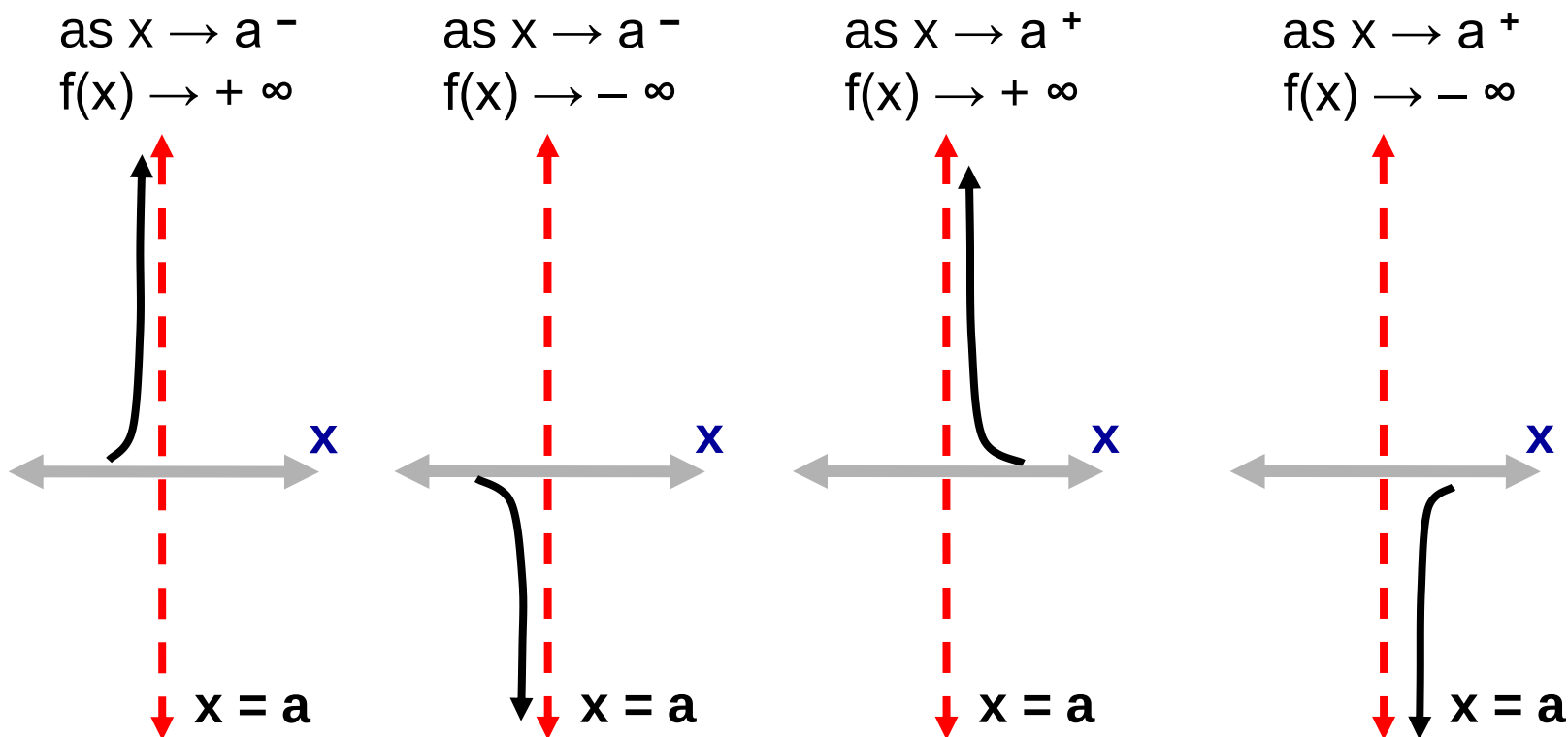
x	$f(x)$
-2	-0.5
-1	-1
-0.5	-2
-0.1	-10
-0.01	-100
-0.001	-1000

as $x \rightarrow 0^-$, $f(x) \rightarrow -\infty$.

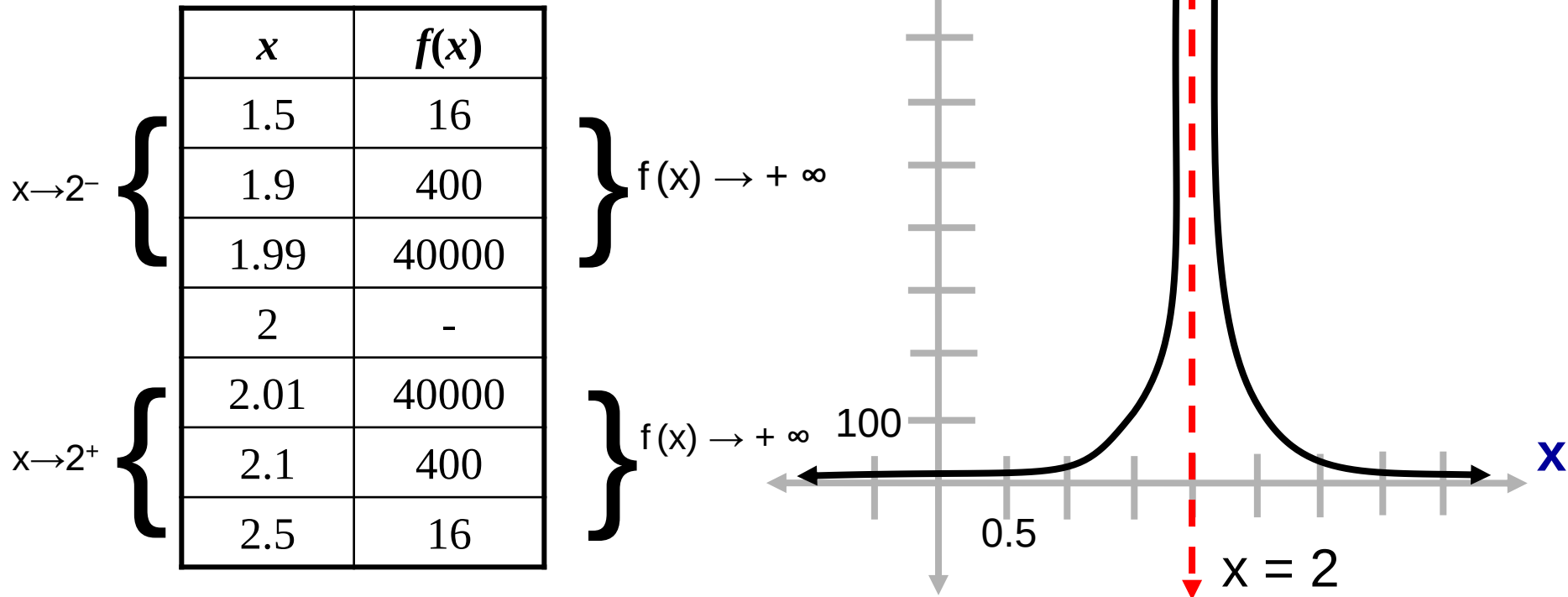


Vertical Asymptote

The line $x = a$ is a vertical asymptote of the graph of $y = f(x)$, if and only if $f(x) \rightarrow +\infty$ or $f(x) \rightarrow -\infty$ as $x \rightarrow a^+$ or as $x \rightarrow a^-$.



Example: Show that the line $x = 2$ is a vertical asymptote of the graph of $f(x) = \frac{4}{(x-2)^2}$



This shows that $x = 2$ is a vertical asymptote.

A rational function $\frac{P(x)}{Q(x)}$ may have a vertical asymptote at $x = a$ for any value of a such that $Q(a) = 0$.

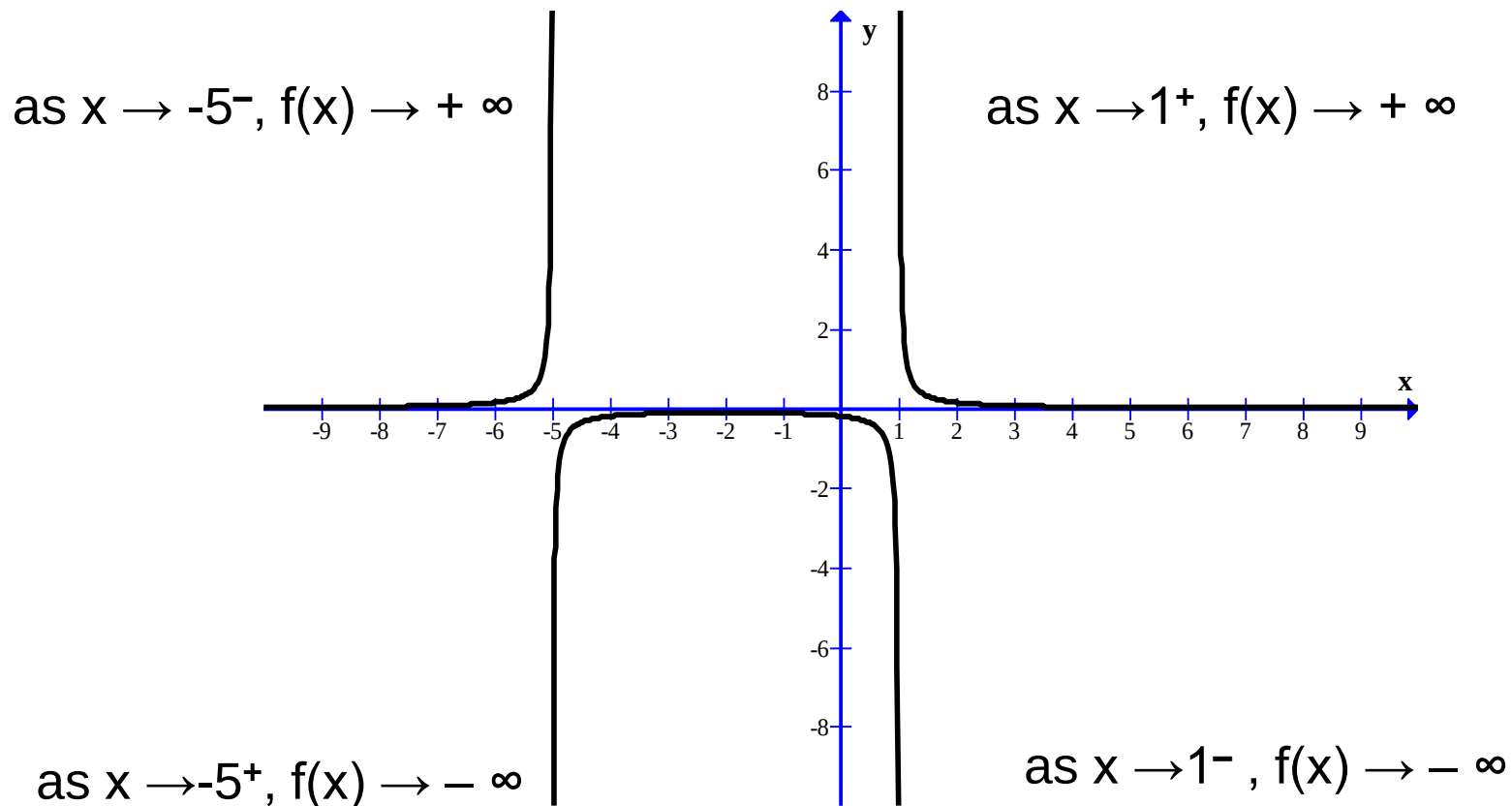
Example:

Find the vertical asymptotes of the graph of $f(x) = \frac{1}{(x^2 + 4x - 5)}$

Set the denominator equal to zero and solve.

Solve the quadratic equation $x^2 + 4x - 5 = 0$ $(x - 1)(x + 5) = 0$

Therefore, $x = 1$ and $x = -5$ are the values of x for which $f(x)$ may have a vertical asymptote.



$x = -5$ is a vertical asymptote.

$x = 1$ is a vertical asymptote.

Horizontal Asymptote

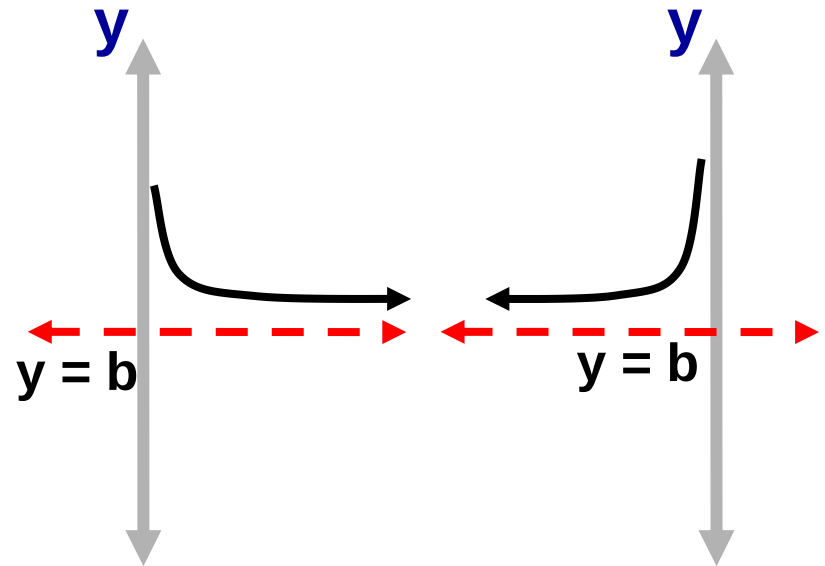
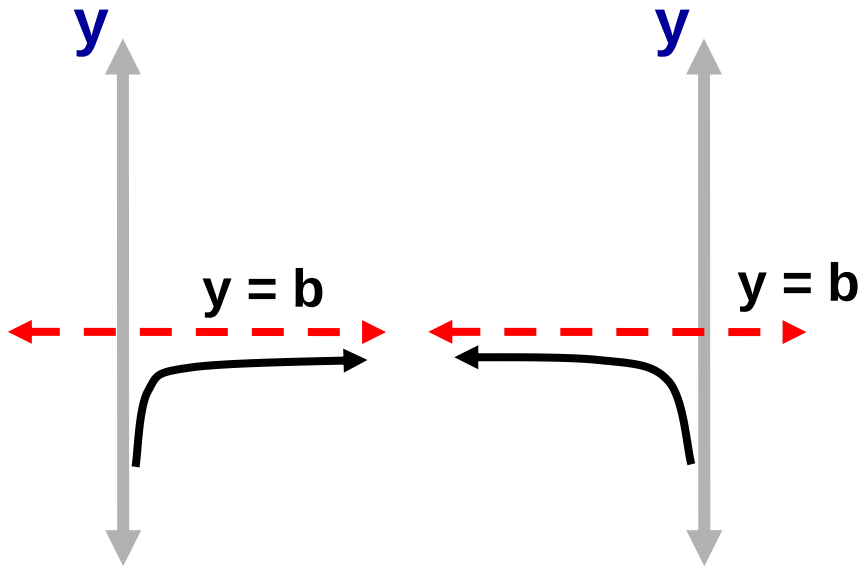
The line $y = b$ is a horizontal asymptote of the graph of $y = f(x)$ if and only if

$$\begin{aligned} \text{as } x \rightarrow +\infty \\ f(x) \rightarrow b^- \end{aligned}$$

$$\begin{aligned} \text{as } x \rightarrow -\infty \\ f(x) \rightarrow b^- \end{aligned}$$

$$\begin{aligned} \text{as } x \rightarrow +\infty \\ f(x) \rightarrow b^+ \end{aligned}$$

$$\begin{aligned} \text{as } x \rightarrow -\infty \\ f(x) \rightarrow b^+ \end{aligned}$$



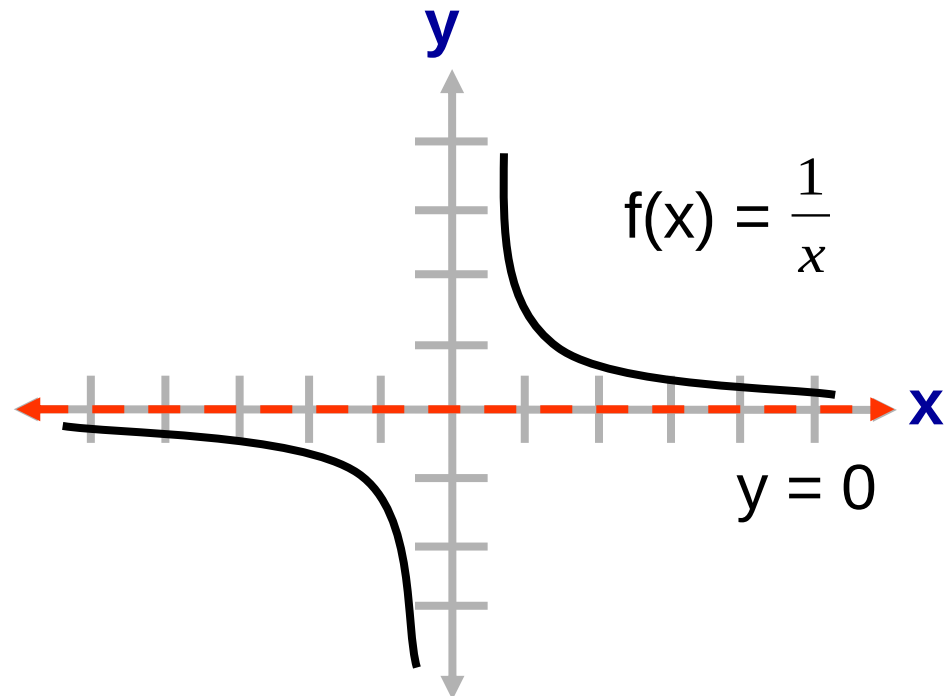
Example: Show that the line $y = 0$ is a horizontal asymptote of

the graph of the $\frac{1}{x}$ -function $f(x) =$

As x becomes unbounded positively, $f(x)$ approaches zero from above; therefore, the line $y = 0$ is a horizontal asymptote of the graph of f .

As $f(x) \rightarrow -\infty$, $x \rightarrow 0^-$.

x	$f(x)$
10	0.1
100	0.01
1000	0.001
0	—
-10	-0.1
-100	-0.01
-1000	-0.001



Nobody wants to plot points to determine a horizontal asymptote!

You can determine the horizontal asymptotes by remembering just 3 little things:

1. If the degree of the num $<$ degree of the denom,
then $y = 0$ is a HA.
2. If the degree of the num $>$ degree of the denom,
there is NO HA.
3. If the degree of the num $=$ degree of the denom,
then the line.

$$y = \frac{\text{LC of num}}{\text{LC of denom}} \text{ is the HA.}$$

Finding Asymptotes for Rational Functions

Given a rational function: $f(x) = \frac{N(x)}{D(x)}$

- Check to factor and cancel.
- Vertical asymptotes occur at x values that make the denominator equal zero.
 - If $n < d$, then $y = 0$ is a horizontal asymptote.
 - If $n > d$, then there are no horizontal asymptotes.
 - If $n = d$, then $y = \frac{LC_m}{LC_n}$ is a horizontal asymptote.

Example:

Find all horizontal and vertical asymptotes of $f(x) = \frac{3x^2 + 6x + 3}{x^2 - x - 2}$

Factor the numerator and denominator.

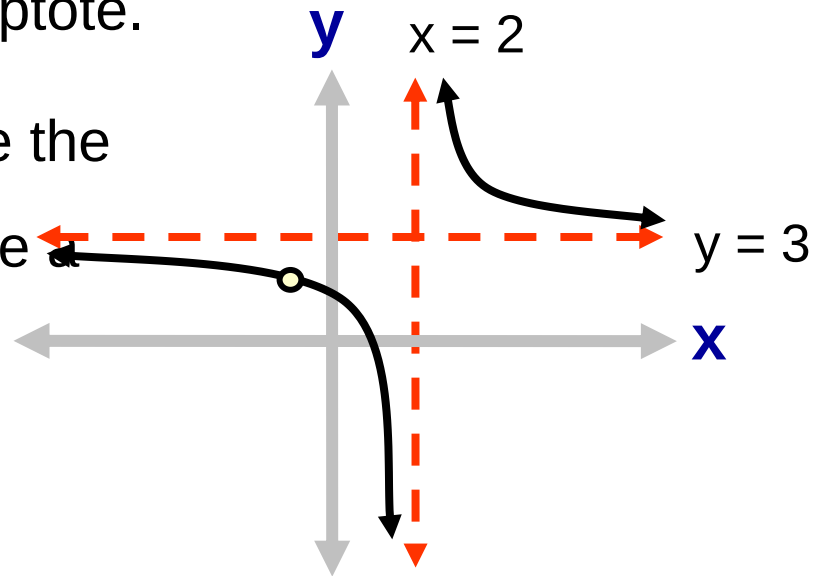
$$\frac{3(x+1)(x+1)}{(x-2)(x+1)}$$

Since $(x + 1)$ will cancel, there is a hole in the graph at $x = -1$.

$x = 2$ will be a vertical asymptote.

Since the polynomials have the

same degree, $y = 3$ will be a horizontal asymptote.



Slant Asymptote

A **slant asymptote** occurs when the degree of the numerator is exactly 1 more than the degree of the denominator.

A **slant asymptote** is found by using long division.

Example: Find the slant asymptote for $f(x) = \frac{2x^2 + x - 1}{x + 3}$

Divide: $\frac{2x^2 + x - 1}{x + 3} = 2x - 5 + \frac{14}{x + 3}$

As $x \rightarrow +\infty$, $\frac{14}{x + 3} \rightarrow 0^+$.

As $x \rightarrow -\infty$, $\frac{14}{x + 3} \rightarrow 0^-$.

Therefore as $x \rightarrow \pm\infty$,

$f(x)$ is more like the line $y = 2x - 5$.

The slant asymptote is $y = 2x - 5$.

